Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Project 2 will help a lot because its very related to the project I chose. Making a little RSA program last year will also help me think about what I need.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect getting the bare fundamentals of the page working to be the easiest, just not making it pretty. Understanding the code for some of the encryption algorithms will probably be the difficult part, just because I want to know why it works more than I need to implement it correctly.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Trouble shooting the crypto algorithms will probably be the most time consuming, but

they also kind of need to be the first ones implemented.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Studying for other classes and completing final assignments will probably interfere the most.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**It will take input of a message, algorithm, and a key. It will return an encrypted message and the key that will decode it on the website, as well as saying which algorithm was used.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	class state: 1: message: list[Message] 2: algorithm: str 3: random: int 4: base: int
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	class Message: plaintext: str cyphertext: str encrypt: bool
Routes (minimum 7) The names of your route functions	● main ● RSA ● ceasar ● MD-5 ● AES ● SHA-256

	• tampered
Pages (minimum 5) A brief description of the pages in your site	• main ○ • RSA ○ • ceasar ○ • MD-5 ○ • AES • SHA-256 ○ • tampered ○
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox: to enter message SelectBox: to select algorithm TextBox: to enter expected hash for tampered page part
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1: to determine which algorithm to use when on the main page when Button clicked 2: like in project 2 for the Ceasar cypher, making sure the characters are printable 3: To see if the message was tampered with in the for the algorithms its implemented in
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in	Basically every page will have one to actually encrypt or decrypt the message(s).

your State attribute. Description of this loop and why it is meaningful to the site	

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

N/A

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.